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Active compliance control of robot peg-in-hole assembly based on combined reinforcement learning

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Abstract
Robot peg-in-hole assembly has always been a research hotspot. With the application of reinforcement learning in the field of robotics in recent years, the assembly strategy based on reinforcement learning has gained significant attention. However, the application of reinforcement learning in physical robots is still limited due to its requirement of a large number of training episodes to explore the environment. In this paper, a novel combined reinforcement learning method is proposed. Firstly, a state judgment Deep Q-Network (SDQN) and a parameter optimization Deep Q-Network (PDQN) are integrated into a framework to train a variable impedance controller. The SDQN strategy can judge the contact state and then select the parameter range for optimization. Then, the PDQN module can generate parameters of the variable impedance controller. Secondly, a reward function for SDQN is designed by incorporating the posture error and insertion depth into the calculation of the negative reward value to ensure assembly speed. The experiment is conducted with a clearance of 0.40 mm, resulting in a success rate of 96% for the assembly task. These findings validate the effectiveness of the proposed combined reinforcement learning method.

Keywords Peg-in-hole assembly · Reinforcement learning · Impedance control · Compliance control

1 Introduction

Industrial robotics have been widely used in pick-and-place, palletizing, spraying, welding, and other low-contact tasks, significantly enhancing work efficiency and intelligence. With the diversification of application scenarios and the advancement in automation technology, industrial robotics need to perform complex tasks in contact-rich unstructured environments. Assembly is a crucial stage in product manufacturing, and assembly performance directly affects product quality [1]. The insertion operation during assembly is a typical contact-rich robotic task, which can be abstracted as peg-in-hole. The assembly operation is usually divided into two subtasks: the searching phase and the inserting phase [2–4]. In the searching phase, a visual servo [5] or a trajectory-based spiral search strategy [6] is typically used to bring the peg close to the hole. In the inserting phase, the robot needs to adjust the motion trajectory according to the feedback force generated by the peg-in-hole contact to control the assembly force. This adjustment process is commonly referred to as compliance control.

This study focuses on compliance control in the inserting phase of the peg-in-hole task. Industrial robots are typically rigid, position-controlled machines, and even a tiny pose error can generate significant contact force during the assembly process, potentially causing task failures or robot damage. To complete the assembly task successfully, the robot needs to exhibit compliance in the insertion phase. Robot compliance refers to the robot's flexibility and adaptability during interaction with external objects or environments, enabling it to adjust its movements and forces flexibly in contact-rich, unknown, or variable environments to complete tasks without exerting excessive force or causing damage. The most commonly used compliance control methods can be categorized as passive compliance control methods and active compliance control methods.

The passive compliance control method exploits the elastic deformation of components installed at the end of the robot to adjust the motion of the workpiece under assembly force. For example, Whitney et al. [7] first analyzed the geometric and force balance conditions required for the

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successful mating of rigid peg-in-hole parts, and based on this, they designed a Remote Center Compliance (RCC) structure with spring suspension and applied it to robotic peg-in-hole assembly tasks. This structure provided the foundation for many passive compliance methods. However, as the application for robots gradually expands to unstructured environments, assembly robots must have the ability to independently explore, perceive, decide, and plan. This places higher demands on the intelligence of robot assembly technology.

With the development of computer and sensor systems, the active compliance control assembly strategy with autonomous exploration and decision-making ability has gained significant attention. Active compliance control enables the robot to correct its motion path through feedback signals, such as force signals or image signals. Active compliance strategies can be mainly divided into model-based strategies and learning-based strategies.

An important method of model-based strategy is to establish explicit equations between environmental characteristics or contact force and axial attitude. Gai et al. [8] discovered significant time-varying and coupling characteristics in the modeling process of the peg-in-hole insertion task and developed a piecewise strategy and a decoupled control (PSDC) method to improve the pose accuracy between the peg and the hole. Park et al. [9] analyzed four contact states of the peg-in-hole insertion task.

Model-based strategies often have high labor costs. Researchers prefer to utilize assembly actions as a skill, enabling robots to formulate assembly strategies through appropriate skill selection and combination. Consequently, researchers have shifted their focus towards assembly strategies based on operation skill learning, including imitation learning and reinforcement learning. Imitation learning is a non-programming strategy that can generate trajectories quickly. Ding et al. [10] developed an unmarked visual capture system based on the Kinect FORTH system to track continuous human hand centroid movements. They utilized the GMM-GMR method to derive optimal trajectories containing sufficient human characteristics and employed Dynamic Motion Primitive (DMP) for trajectory generalization learning. Finally, an improved Visual Spatial Skill Learning (VSL) algorithm was adopted to enable the robot to learn target configurations directly from a multi-operation demonstration video, reflecting the actual pick-and-place points for the nail-in-hole task. However, this vision-based imitation learning method requires a wide-open and unobstructed assembly platform, which deviates greatly from practical applications. Song et al. [11] divided the assembly process into five stages and used the Gaussian mixture model (GMM) to achieve compliance control. Liu et al. [12] used an improved DMP model with a novel hybrid force/position feedback in Cartesian space for the insertion phase of a robotic peg-in-hole assembly task. The modified DMPs generated trajectory is tracked using the Cartesian impedance control method. The model reduces trajectory errors quickly and effectively by introducing force and position feedback, resulting in smooth robot behavior and active posture adjustment to avoid jamming. Similar to human learning assembly operations, it is difficult to obtain and comprehend assembly skills only from the general behaviors demonstrated by experts, and the collection and processing of expert data remains a significant challenge. The assembly strategy based on reinforcement learning (RL) provides a potential solution to address these issues. The robot assembly strategy based on RL is an intelligent algorithm, which adopts the idea of RL to enable the robot to learn gradually by interacting with the environment and determining which actions to take in different states to maximize cumulative rewards. In peg-in-hole assembly tasks, the robot needs to be able to independently explore and perceive, judge the state information such as the position of components and the robot's posture, and select appropriate actions, such as moving the robotic arm and rotating tools, according to the current state to complete the assembly task. Through training with RL algorithms, the robot can improve its assembly capability over time and thus complete peg-in-hole assembly tasks efficiently and accurately, thereby providing a new technological approach for automation in production lines. As a robot assembly strategy, RL only requires a small amount of prior knowledge and human involvement and it provides a reliable method for robot skill learning.

This paper proposes robot peg-in-hole assembly active compliance control based on combined reinforcement learning (CRL). The main contributions of this method are as summarized follows:

(1) A novel RL-based assembly method is proposed, which combines two DQN models for training. The method utilizes a state judgment DQN (SDQN) and a parameter optimization DQN (PDQN) to train a variable impedance controller for controlling contact forces during the assembly process. The damping parameters of the impedance controller are continuously optimized to ensure a fast and stable assembly process.
(2) A new subsection-based parameter optimization strategy is introduced. The SDQN evaluates the contact state and obtains a parameter optimization strategy, which is then utilized by the PDQN to optimize the damping matrix based on the evaluation from the SDQN. Instead of exploring a wide range of parameter values, this strategy adopts a subsection approach and selects the parameter range for optimization according to the contact state, ensuring a fast and stable assembly process.

(3) A reward function for the SDQN is proposed, which incorporates the posture error and insertion depth into the calculation of the negative reward value. This reward function guarantees both the assembly speed and the quality of the insertion task by ensuring the position and posture accuracy of the peg during the insertion task.

The rest of this paper is organized as follows. Section 2 reviews related literature. Section 3 describes and analyzes the assembly strategy and proposes a compliance control method based on combined reinforcement learning. Section 4 outlines the construction and verification of the experimental platform based on the UR5 robot. Finally, Section 5 presents the conclusion and future work.

2 Related works

At present, when using real robot systems with RL, the common method is to learn the high-level control strategy of the manipulator and design an underlying controller to execute the high-level commands of the RL agent. For instance, Beltran-Hernandez et al. [13] obtained the errors of the peg at the current position and the target position through a camera and combined them with the force information of the insertion process processed by the time convolutional neural network (TCNs) to form the input of the RL algorithm. However, the workspace in an industrial environment is often narrow, making it difficult to obtain a complete six-dimensional attitude of the end effector through a camera in the entire assembly process. The force/position controller can be utilized to satisfy the need for environment contact in the insertion task. Li et al. [14] represented the quaternion sequence of the assembly process within the Markov decision framework. The input of the RL model is the contact state of the assembly process, and the output is an action sequence that combines the movement and rotation of the robotic end-effector in the X, Y, and Z directions. Ren et al. [15] employed deep deterministic policy gradient (DDPG) to learn compliance control along the trajectory in the peg-in-hole task, where the motion space consists of basic motion instructions and stiffness. However, due to the non-heuristic action selection and unlimited random exploration in this force/position controller, the RL algorithm requires many datasets for training.

The general force/position controller divides the task space into the position control subspace and the force control subspace. Only the position of the manipulator is controlled in the position control subspace, while the contact force with the environment is controlled in the force control subspace, and the dynamic coupling between the manipulator and the environment is ignored. In contrast, impedance control considers the dynamic relationship between various variables of the manipulator, such as the position and the force of the end-effector, rather than controlling these variables separately. This allows for simpler and more effective high-precision assembly tasks [16], making the combination of impedance control and RL more straightforward and convenient. Buchli et al. [17] proposed an RL algorithm based on PI2 (the improved path integral strategy) and applied it to realize variable impedance control in robotic operation tasks. They applied the concept of adaptive impedance to PD negative error feedback control in robotics and adopted PI2 reinforcement learning to find a suitable gain plan for a given task. Bogdanovic et al. [18] investigated the influence of parameterizing the output strategy on the RL task of rigid robotics with complex contacts. They learned a strategy of output impedance and desired position in the joint space by using the DDPG algorithm. These studies demonstrate that when the contact state changes dynamically, using the variable impedance controller parameters as the action space of RL provides an effective scheme to reduce the contact force. The variable impedance control in the end-effector space (VICES) can serve as a favorable action space for RL, where the impedance gain of each axis in the six axes (rotation and shift) can be dynamically adjusted according to different states of tasks. This approach enhances sampling efficiency and has a low energy consumption [19], [13].

However, properly adjusting the controller parameters for the robotic peg-in-hole assembly task remains a challenge. In high-precision assembly tasks, even a small numerical change can result in task failure. The value-based RL algorithm has been proven to be effective in dynamically outputting controller parameters during the assembly process. Hou et al. [20] developed an RL-based genetic evolution algorithm to optimize impedance controller parameters and complete the dual peg-in-hole assembly task successfully. Oikawa et al. [21] proposed a variable impedance controller for local trajectory planning based on a non-diagonal stiffness matrix. The contact state is determined using the Deep Q-network (DQN) algorithm based on the contact force, and the insertion task is solved considering the different characteristics of the non-diagonal components on the stiffness matrix. Zou et al. [22] applied the fuzzy Q-learning algorithm to adopt a variable impedance controller in the insertion phase. This approach generates compliant behaviors of the robot during the hole insertion process, improves adaptability, and enhances robustness in a new state or environment. However, it may not be suitable for high-precision assembly tasks. Additionally, a large range of random parameter searches in the training process can result in low learning efficiency, while a small range of parameters may fail to achieve optimal strategy and reduce assembly quality. Table 1 summarizes the characteristics and test results of several RL algorithms in peg-in-hole assembly.

Table 1 Overview of RL algorithms in peg-in-hole assembly

Algorithm	State	Action	Episodes	Training time (min)	Characteristics	Maximum assembly force (N)/ Torque (N·m)	Assembly time (s)	Clearance (mm)	Success rate
Deep Q-learning based on FA [14]	Force, Torque (6 states)	The rotation angles along X, Y, and Z (6 actions)	1000	900	Applicable to the assembly of complex peg-in-hole parts without the need for prior knowledge.	200/6	0.6		84.2%
Fuzzy Q-Learning [22]	Torque, Position, Orientation (12 states)	Damping parameter (3 actions)	160		Train fast and exhibit good robustness during the insertion phase.	40/4		1	100%
Double DQN [23]	Force, Torque (6 states)	Force/torque command action (6 actions)	200	250	A force-based RRRL approach has been developed, which restricts the contact forces generated during the assembly process within a very small safety range.	15/-	2.3		95%
rLfD [24]			700	300	Combine DMP with RL, and adapt to different geometric shapes through a few-shot task.	10/-	6.5	0.4	87%
DDPG [25]	Position, Orientation, Joint angles, Force, Torque, Hole position and direction, Safety status, RGB image, Depth image (27 states)	The translation and rotation of X, Y, and Z (6 actions)	5000		The end-to-end assembly process is performed, and training is conducted using a digital twin model. As a result, a high assembly success rate is achieved.	30/-		1	90%

In this study, an improved value-based RL algorithm is proposed, which combines traditional active force control strategies and RL optimization algorithms. The impedance controller is used to ensure the safety and stability of the assembly task, while the improved DQN algorithm is used to optimize and improve the efficiency of the assembly task. The method uses a segmented strategy to explore and optimize the parameters of the impedance controller, enabling it to have good adaptability during the robot's execution of the axle-hole assembly insertion task, greatly enhancing its assembly efficiency.

3 Peg-in-hole assembly strategy based on combined reinforcement learning

3.1 Method framework

The structure of the combined reinforcement learning (CRL) method proposed in this study for peg-in-hole assembly compliance control is presented in Fig. 1. The Markov decision process (MDP) can be defined by the tuples (S, A, T, R, γ), where S represents a set of states, A represents a set of actions, and transition probability T: $S\times A\times S\rightarrow[0,\infty)$ represents the probability density from $S_t\in S$ to $S_{t+1}\in S$ when taking action $a\in A$. The reward function R: $S\times A\rightarrow R$ determines the reward for each transition, and $\gamma\in[0,1)$ is the discount factor. In this work, there are two independent MDPs. The MDP M^s of the state judgment module is defined by the tuples (St, $Stage$, r^s, T^s, γ^s), where the transition probability T^s represents the probability density from $S\in St$ to the next state $S_\in St$, r^s is the reward function that represents the reward of each transition. Similarly, the MDP M^q of the parameter optimization module is defined by the tuples (S, A, r^q, T^q, γ^q), where the transition probability T^q represents the probability density from $S^q\in S$ to the next state $S^q_\in S$ after performing action A, and r^q is the reward function that maps the current state S and current action to the actual value reward. The state S in this strategy is determined by St and the action $Stage$ of MDP M^s. RL can be used to learn an optimal state judgment strategy π^s: $St\rightarrow Stage$ and a parameter optimization strategy π^s: $S\rightarrow A$ to maximize the expected reward r^s and r^q. The meanings of some symbols in Fig. 1 are listed in Table 2. The dashed arrow on the far right in Fig. 1 indicates that Sq_ is not directly obtained after executing the action, but it is obtained through a combination of the S_ and the action output of the Q-Network of SDQN.

The framework consists of a state judgment DQN (SDQN) module, a parameter optimization DQN (PDQN) module, and an impedance control module. Specifically, the SDQN module and the PDQN module form the CRL, which enables state judgment and parameter optimization functions, respectively. The action spaces of SDQN and PDQN correspond to contact state judgment and variable impedance controller parameters. The SDQN evaluates the contact state of the peg-in-hole assembly task based on the feedback signals from the robot and sensor; then, it selects the appropriate parameter optimization strategy, which is utilized by the PDQN to optimize the impedance parameters of the variable impedance controller. These parameters are

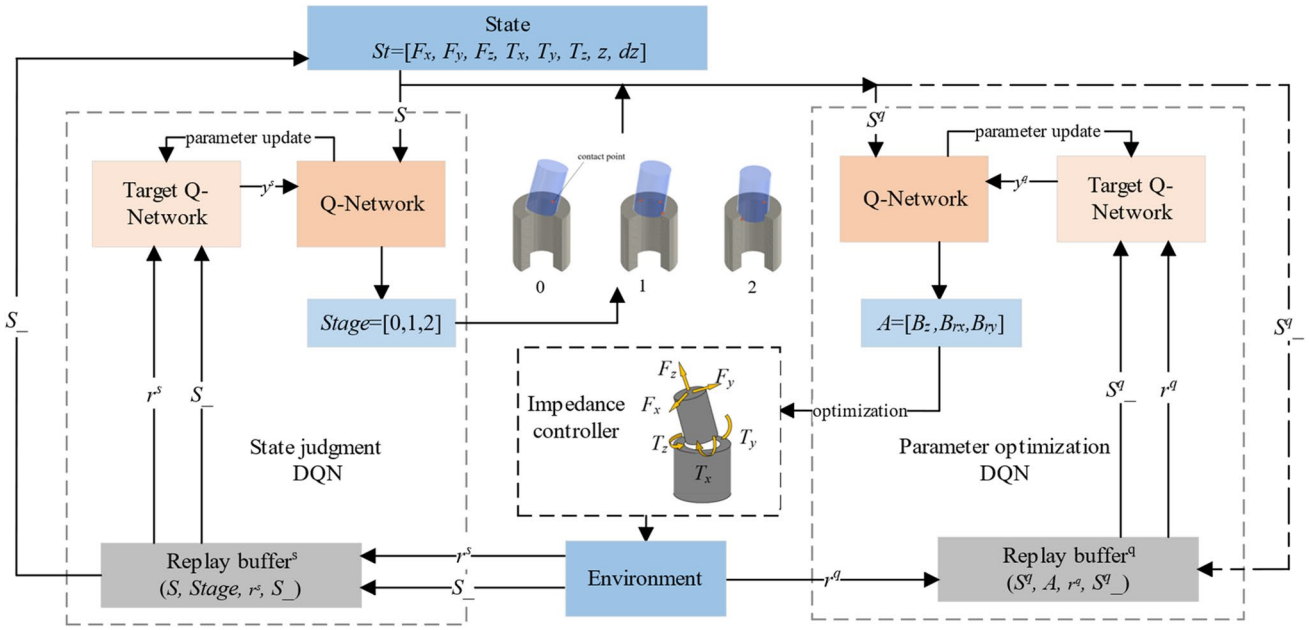


Fig. 1 The schematic of the CRL method

Table 2 The meanings of some symbols in the CRL

Symbols	Meaning
S	The state input of the state judgment module, which is obtained from the force sensor response and the robot position response
S^q	The state input of the parameter optimization module
y^s, y^q	Q value of the SDQN and the PDQN target network
r^s	Reward of SDQN
r^q	Reward of PDQN
$S_$	Observation of SDQN after interacting with the environment
$S^q_$	Observation of PDQN
A	Three-dimensional impedance parameters output by the PDQN

subsequently transmitted to the impedance control module. The impedance control module directly outputs the robot motion commands to guide robotic movement, thereby achieving active compliance control in peg-in-hole assembly.

The dynamic behavior of the robot can be described as a system with mass, damping, and stiffness. The purpose of impedance control is to manage the relationship between force and displacement of the robot, enabling stable interactions between the robot and its environment. This study achieves impedance control in a 6-DOF Cartesian space at the center of the robotic tool. The peg-in-hole assembly task involves a nonlinear dynamic process. The stiffness inertia and damping characteristics within the impedance control system are crucial factors that determine the force-motion relationship. In robot impedance control, the influence of stiffness can lead to tracking error and elastic resilience force, which are not favorable for robotic high-precision position control tasks [26]. Moreover, in the peg-in-hole assembly task, the stiffness of the system can be sufficiently high to be disregarded. Therefore, the selected impedance control law in this paper can be represented by:

$$F_d - F = M_d\left(\dot{V}_d - \dot{V}\right) + B_d\left(V_d - V\right) \tag{1}$$

where M_d and B_d are the diagonal positive definite matrices representing the system inertia and damping, respectively. V_d and V denote the desired and actual Cartesian velocity vectors at the end of the robot, respectively. F_d represents the desired external force vector, while F represents the measured actual external force vector at the end of the robot.

In robot impedance control, the inertial diagonal matrix M_d is often considered constant as it represents the system mass. In this study, the diagonal elements of the diagonal damping matrix B_d serve as the optimization objective for PDQN. By adjusting B_d, this study obtains the desired robot motion through the contact force and modulates the conversion between the force error signal and the velocity signal to achieve compliance control in the insertion phase.

The CRL method proposed in this study is based on the conventional speed controller integrated with the robot, requiring no modification to the original control system. So, it is convenient and straightforward to implement on industrial robots. This method trains the SDQN and PDQN concurrently in the insertion phase. SDQN and PDQN are two interconnected RL systems, each with its own independent reward system and experience replay pool. Detailed introductions to them will be provided in Sections 3.2 and 3.3 respectively. The training process is outlined in Algorithm 1.

3.2 SDQN module

(1) Contact state analysis of peg-in-hole insertion: In the contact-rich peg-in-hole assembly task, the progress of the task is often accompanied by a change in the contact state. Qiao et al. [27] regarded mechanical assembly as a process from the current point to the target point. They analyzed the contact state of several peg-in-hole assembly tasks with different shapes in three-dimensional space and identified a transition

Algorithm 1 The CRL training algorithm

```
Initialize the Q-Network and the target Q-Network of two DQNS with random weights.
Initializes the replay buffer R^s and R^q for SDQN and PDQN.
for episode = 1 to M do
    Reset the environment and obtain the initial state S of SDQN.
    SDQN selects the action Stage according to S.
    The initial state S^q of PDQN consists of S and Stage.
    for step = 1 to T do
        PDQN selects action A according to S^q.
        Execute action A and receive the rewards rs and rq; obtain the next state S_ of SDQN.
        Store transition (S, Stage, r^s, S_) in R^s.
        SDQN selects the action Stage_ according to S_.
        The next state S^q_ of PDQN consists of S_ and Stage_.
        Store transition (S^q, A, r^q, S^q_) in R^s.
        Update the four networks.
    end for
end for
```

state called the three-point contact state in the insertion phase of peg-in-hole assembly. Our study divides the peg-in-hole insertion process into three states: single-point contact, three-point contact, and two-point contact. The three contact states represent different insertion processes. As the insertion task continues, the posture error becomes smaller, and the allowable range of posture adjustment is reduced. Therefore, a large expected impedance parameter value should be selected within the safety range to ensure that the assembly task is completed safely in the case of sudden contact force in a narrow space.

(2) SDQN: In the precision peg-in-hole assembly task, it is difficult to model the relationship between the required motion of the robot and the assembly contact force/torque. Meanwhile, the robot peg-in-hole assembly is a highly real-time control task. Oikawa et al. [21] pointed out that constructing a contact point model through explicit equations may not provide the action output of the model in a short period of less than tens of milliseconds. In contrast, the DQN reinforcement learning method can judge the contact state and select the assembly action more quickly. Therefore, this study adopts the DQN model to determine the single-point contact, three-point contact, and two-point contact states. At the beginning of one episode, the initial state S is defined by the contact between the robot and the environment. S is an eight-dimensional vector composed of six-dimensional force information, and S is defined as:

$$S = [F_x, F_y, F_z, T_x, T_y, T_z, z, dz] \tag{2}$$

where z represents the depth in the current Z direction, and dz represents the offset in the Z direction in the previous episode. Then, S is input into the Q-network of SDQN, and the output action is a one-dimensional number, with 0, 1, and 2 representing the three states of single-point contact, three-point contact, and two-point contact of the peg-in-hole insertion process.

(3) Reward function: In this paper, the reward function of the SDQN mainly consists of two parts:

$$r^s = r_1^s + r_2^s \tag{3}$$

where r_1^s and r_2^s represent the positive and negative rewards to end a training set, respectively. r_1^s is set to:

$$r_1^s = 1 - k/k_{max} \tag{4}$$

where k represents the current number of steps, and $kmax$ represents the maximum number of steps manually set in a training set in the experiment. r_1^s gives more positive rewards to fewer assembly steps, and its function is to promote the inserting action.

Negative rewards are used to penalize states where assembly actions are stalled. In the insertion phase, the posture error is continuously corrected, and the insertion depth is continuously increased. In reference [11], Gai pointed out that a major problem of existing compliance control methods is that the evaluation indicator of the assembly process is only the force/torque precision, and the pose precision between the peg and the hole cannot be guaranteed. However, the posture of the peg is coupled with the contact force/torque, indicating that the adjustment of posture error between peg-in-holes does not only depend on the feedback force and torque. Therefore, to determine whether the assembly action can be conducted smoothly, the posture adjustment and torque control should be comprehensively considered.

At the beginning of the insertion phase, posture adjustment is often accompanied by a change in the contact state, and one of the functions of the SDQN is to promote the smooth progress of the assembly process. Thus, this study incorporates the posture error and insertion depth into the calculation of the negative reward, and the function of r_2^s is defined as:

$$r_2^s = -1/z - \theta/\cos\theta \tag{5}$$

where z (mm) denotes the current depth in the Z direction, and θ denotes the posture error. The determination of θ is shown in Fig. 2. In this module, a successful posture adjustment is represented by a decrease in posture error and an increase in insertion depth, leading to a larger reward value. In contrast, if an inappropriate assembly strategy is adopted in the insertion process, a failure of the assembly action will cause the SDQN to give a penalty.

3.3 PDQN module

(1) *PDQN*: Through the SDQN module, the contact state can be determined, and the corresponding damping parameter search strategy or range can be obtained. Then, the damping parameters need to be further optimized in the search range to control the contact force within the expected range. Kozlovsk et al. [28] studied the optimal choice of asymmetric damping matrices to improve the ability to perform assembly tasks. Reference [21] developed a variable impedance controller based on the fuzzy Q learning algorithm. This controller replaces the complex Q function with fuzzy rules to optimize the six diagonal parameters of the damping matrix, corresponding to the six motion units of movement and rotation in the X, Y, and Z directions of the robot end-effector's motion space. In this way, compliance behavior is produced during the peg-in-hole inser-

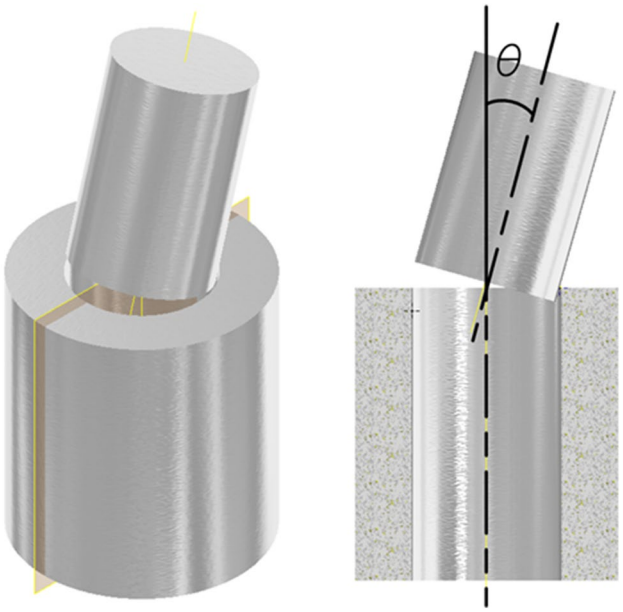


Fig. 2 The posture error θ between the peg and the hole

tion process. The value-based RL algorithm optimizes the diagonal terms of the damping matrix, and it is an effective approach in peg-in-hole assembly compliance control. This study uses PDQN to optimize the diagonal terms of the damping matrix of the variable impedance controller. The state input of the PDQN is a nine-dimensional vector S^q composed of S and *Stage*, and it is defined as:

$$S^q = [F_x, F_y, F_z, T_x, T_y, T_z, z, dz, Stage] \tag{6}$$

For the selection of PDQN action space, this study first considers the damping parameters in R_x and R_y directions that have the greatest impact on assembly action. The general impedance control model often ignores the impact of the constantly changing friction force on the system in the Z direction of the axis, but Xia et al. [29] pointed out by establishing a dynamic model based on compliant contact mechanics that the influence of friction may cause the model output to be inconsistent and ambiguous, resulting in invalid assembly actions, consuming more assembly time, and violating the principle of energy conservation. Therefore, the damping parameter in the Z direction is regarded as the scope of parameter optimization in this study. Therefore, the Q-Network of PDQN outputs a three-dimensional action sequence A. The three elements in A are used as damping coefficients in Z, R_x, and R_y directions respectively, and thus action A is expressed as:

$$A = [B_z, B_{rx}, B_{ry}] \tag{7}$$

According to the three contact states, the optimization range of the three damping coefficients is defined in advance, i.e., each contact state corresponds to a different parameter optimization strategy. The three elements in A are used to update the damped diagonal positive definite matrix in the impedance control. The impedance controller manipulates the robot to generate actions, interact with the environment, generate two DQN reward values (r^s and r^q), and enter a new state at the same time. As shown in Fig. 1, this new state is the next state $S_$ of the SDQN; r^s and $S_$ are stored in its experience replay pool. $S_$ is input to the SDQN, which is combined with the output *Stage*_ to generate the next state $S^q_$ of the PDQN. $S^q_$ and r^q are input into its experience pool and used as the state of the next episode. The episodes are repeated until the assembly task is completed or the maximum number of assembly steps is reached, and at this time, an episode ends. Then, the robot returns to the initial state, enters the next training set, and continues to explore and learn until it obtains the optimal strategy.

(2) Reward function: Similarly, the reward value r^q of the PDQN is also divided into a positive reward r_1^q and a negative reward r_2^q. A larger feedback force and more assembly steps will lead to smaller reward values. Let PDQN's positive reward be $r_1^q = r_1^s$. In contrast, its negative reward r_2^q is calculated by using a fuzzy reward system. The system can consider more factors and set fuzzy rules to describe more complex rewards by embedding the prior knowledge of assembly tasks, instead of designing a complex reward function. The fuzzy reward system combined with prior knowledge evaluates the assembly process. Compared with the previous work, it considers more comprehensive factors, and thus it can improve the learning efficiency and prevent the agent from falling into the local minimum [30].

The fuzzy reward system considers the impact of four factors, including the insertion depth z of the axis, the offset δ_z in the Z direction at this step, the current contact force, and the current torque. The reward system optimizes the parameters in the three directions of Z, R_x, and R_y in the diagonal damping matrix, respectively. The relationship between the damping parameters of each direction and its force/torque input is listed in Table 3.

The PDQN adopts a two-layer fuzzy rule system to represent the membership function. As illustrated in Fig. 3, the total number of fuzzy rules is 75, and each fuzzy set includes 25 rules. This fuzzy reward system is designed based on prior assembly knowledge, and it can be easily adjusted to meet the requirements of different tasks [22]. Different from

Table 3 The correspondence between the input and output in the fuzzy rules

Fuzzy rule input	Damping parameters
F_x, T_x	B^d_{rx}
F_y, T_y	B^d_{ry}
F_z, max(T_x,T_y)	B^d_z

the precise calculation of the function, the input values of the fuzzy set are divided into five triangular affiliations: very bad (VB), bad (B), normal (N), good (G), and very good (VG). Taking the input of the robot position response (z and δ_z) and the force sensor response (F and T), the two-layer fuzzy reward system can judge its membership degree through the performance of different phases in the assembly process and obtain the importance weights of the member relationship in this state. Finally, the function of outputting a negative reward r_2^q is used to evaluate the state obtained after the action is executed. The final output r_2^q of the two-layer fuzzy reward system is presented in Table 4.

Table 4 The reward output of the final layer

Depth sets / Force sets	VB	B	N	G	VG
VB	-1.0	-1.0	-0.69	-0.48	-0.48
B	-0.69	-0.69	-0.48	-0.33	-0.33
N	-0.48	-0.48	-0.33	-0.33	-0.23
G	-0.33	-0.23	-0.23	-0.16	-0.16
VG	-0.23	-0.16	-0.16	-0.11	-0.11

4 Experiments

4.1 Experimental platform

This section verifies the proposed CRL method on a six-axis robot. The experimental platform is demonstrated in Fig. 4. The assembly experiment is conducted on a UR5 robot equipped with a six-dimensional force/torque sensor at its end. A two-finger gripper is connected to the robot through a sensor. The main hardware configurations for this experimental platform include a UR5 collaborative robot with an

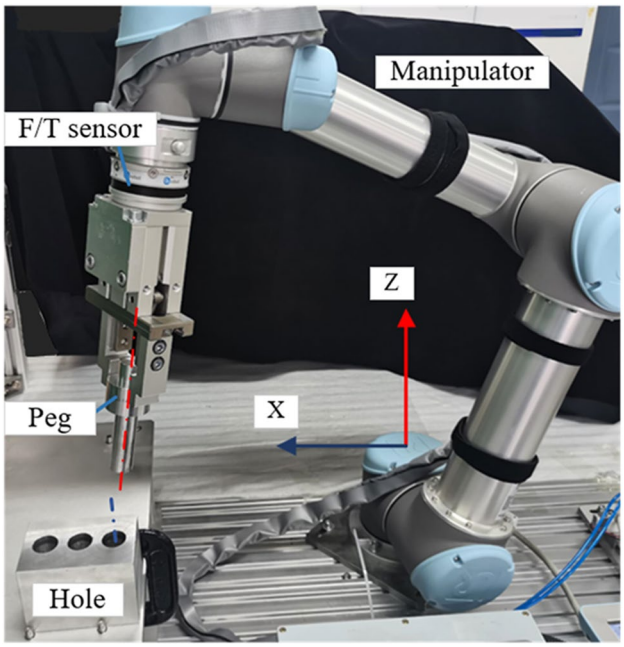


Fig. 4 The UR5 robot platform

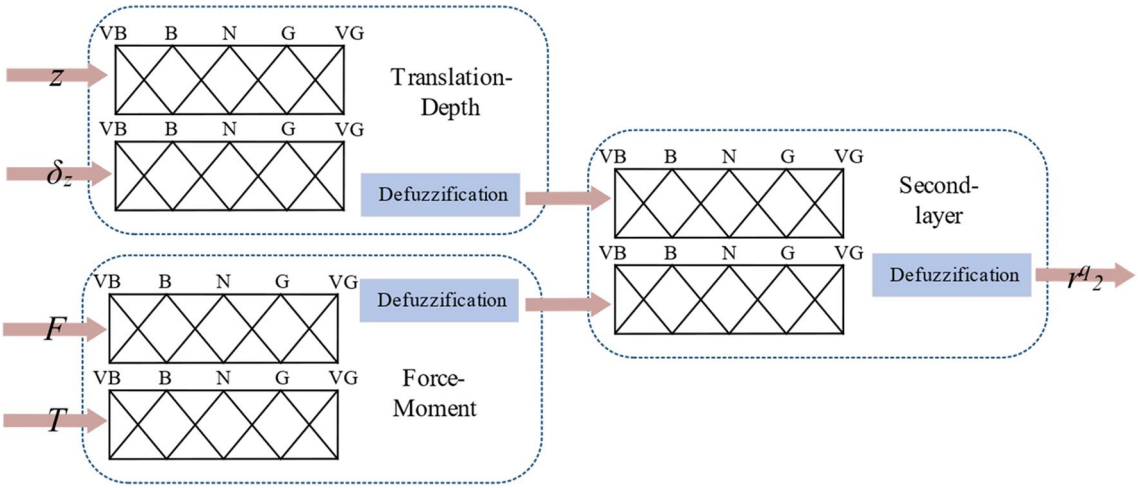


Fig. 3 The two-layer fuzzy reward system

on-robot six-dimensional force/torque sensor. The computer used is equipped with an Intel i5 CPU and a GeForce GTX 1650 graphics card. The software configurations consist of the Ubuntu 18.04 operating system, Python 3.6, and TensorFlow 2.1 framework. Communication between the robot and computer is established through the RTDE protocol, while the force sensor communicates with the computer through the Socket protocol. As shown in Fig. 5, the diameter of the peg is 23 mm. The experiments of peg-in-hole insertion experiments are performed in holes with diameters of 23.4 mm, 23.8 mm, and 24.2 mm, respectively. The contact force/torque is collected in the assembly process and transmitted to the computer. The computer processes the force/torque signals and the three modules of the CRL method to achieve assembly compliance control.

4.2 Insertion phase experiment

This experiment aims to verify the performance of the proposed CRL method under different assembly clearances by attempting to insert the peg into three different holes. To illustrate the experimental process more conveniently, this study describes the training and assembly process under a hole diameter of 23.4 mm.

First, the robotic arm is manually moved to different initial positions in the insertion phase of the assembly task. Then, the insertion phase of the assembly process is started. In this phase, the computer collects the contact force/torque information through the force sensor and responds to the external with the current position information of the robot, which is automatically queried. Meanwhile, the computer continuously transmits the response to the variable impedance controller module of the CRL, thus obtaining robot motion commands to control the assembly force. To ensure the stability of the assembly process, the inertia matrix M_d of the impedance controller in this experiment is set to diag[0.008, 0.008, 0.008, 0.008, 0.008, 0.008], and the damping matrix B_d is set to diag[300, 300, B_z^d, B_{rx}^d, B_{ry}^d, 50], where the damping coefficients B_z^d, B_{rx}^d, and B_{ry}^d in the directions of Z, Rx, and Ry are given by the proposed CRL method.

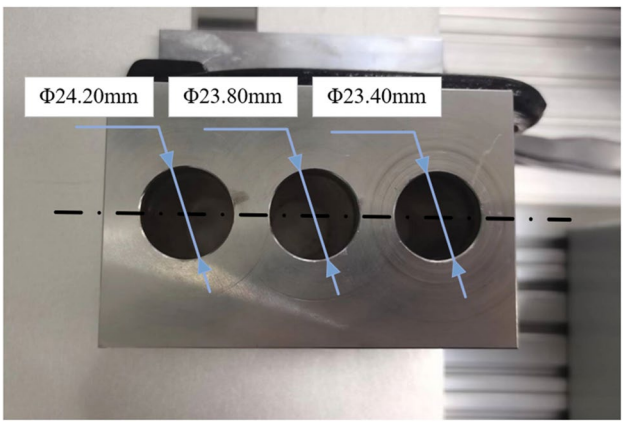
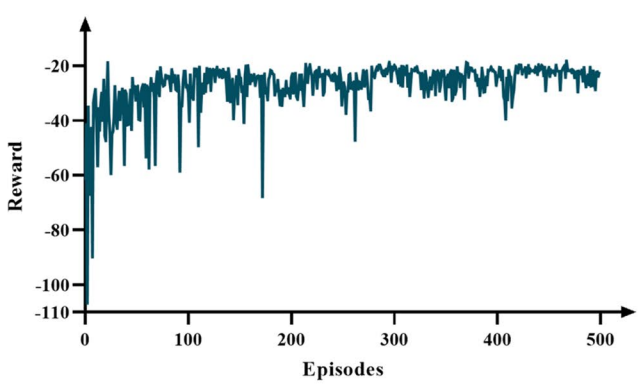


Fig. 5 Three different diameters of holes

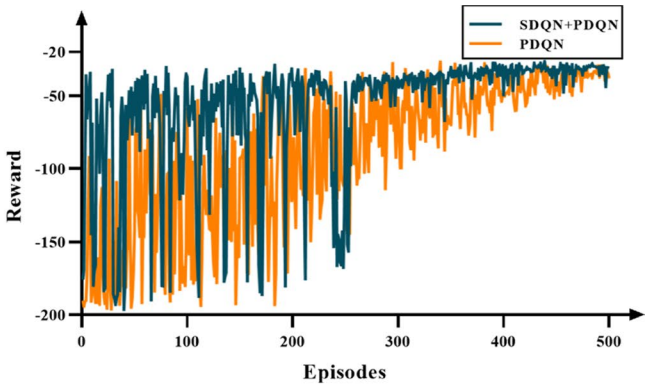
The insertion experiment takes the peg at 5–15 mm above the hole as the initial state. In this experiment, the angle error between the initial state of the peg and the robot base in the X and Y directions is approximate [10.85°, 3.07°]. When the insertion depth is $z = 60$ mm, the assembly task is determined to be completed. The learning rate is set to 0.3. The maximum number of steps per episode k_{max} is 200. First, SDQN + PDQN is compared with a single PDQN in the same learning environment to evaluate its performance. The results are shown in Fig. 6. It can be seen that the CRL method using SDQN + PDQN can converge faster and learn assembly strategies with fewer steps and higher cumulative rewards.

The experimental results indicate that the SDQN converges at around 420 episodes. This demonstrates that the posture error of the insertion task can be eliminated more smoothly as the training process continues, and the SDQN promotes the assembly action. The reward value of the PDQN converges sharply around 260 episodes. This indicates that as the pose error is continuously eliminated, the number of steps required to complete the assembly becomes smaller, and the assembly task can be completed in less time. Moreover, in a single PDQN method, the reward value and steps tend to converge around 500 episodes, while in the CRL method, the reward value converges around 400 episodes. This proves that SDQN has a positive impact on PDQN, and it also demonstrates that the impedance parameters are continuously optimized during the assembly process. In other words, in the changing state during the insertion process, the agent can choose a more appropriate variable impedance strategy to control the assembly force/torque. The entire insertion process is illustrated in Fig. 7. The curves depicting the change in posture error and the trajectory of the robotic end are illustrated in Figs. 8 and 9, respectively. The contact force during the assembly process is presented in Fig. 10, while the torque generated during the same process is illustrated in Fig. 11.

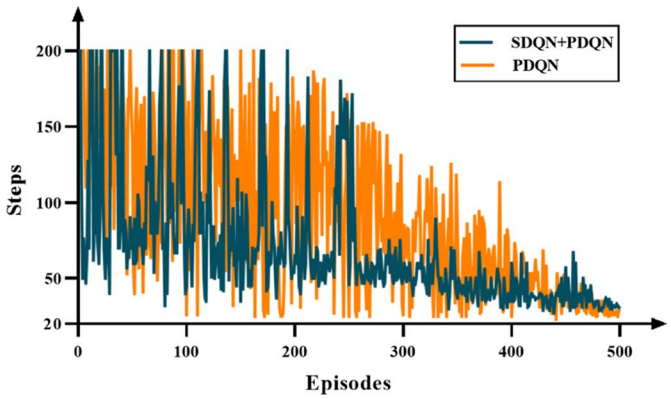
As shown in Figs. 10 and 11, the insertion process is completed in about 17 s. The contact force component in each direction can be kept within 20 N, while the torque component remains stable within 2 N·m. The peg starts to descend along the Z axis, and it contacts the hole at approximately 1.2 s to generate force and torque. Then, the peg-in-hole reaches a single-point contact state. In this stage, there is a relatively large lateral error, and the change in force information is relatively simple. In the insertion phase, the impedance control limits the contact force generated in the X and Y directions to below 4 N, and the contact torque



(a) The reward change curve of the SDQN



(b) The reward change curve of the PDQN



(c) The change curve of the number of assembly steps

Fig. 6 The performance curves of the CRL training insertion task

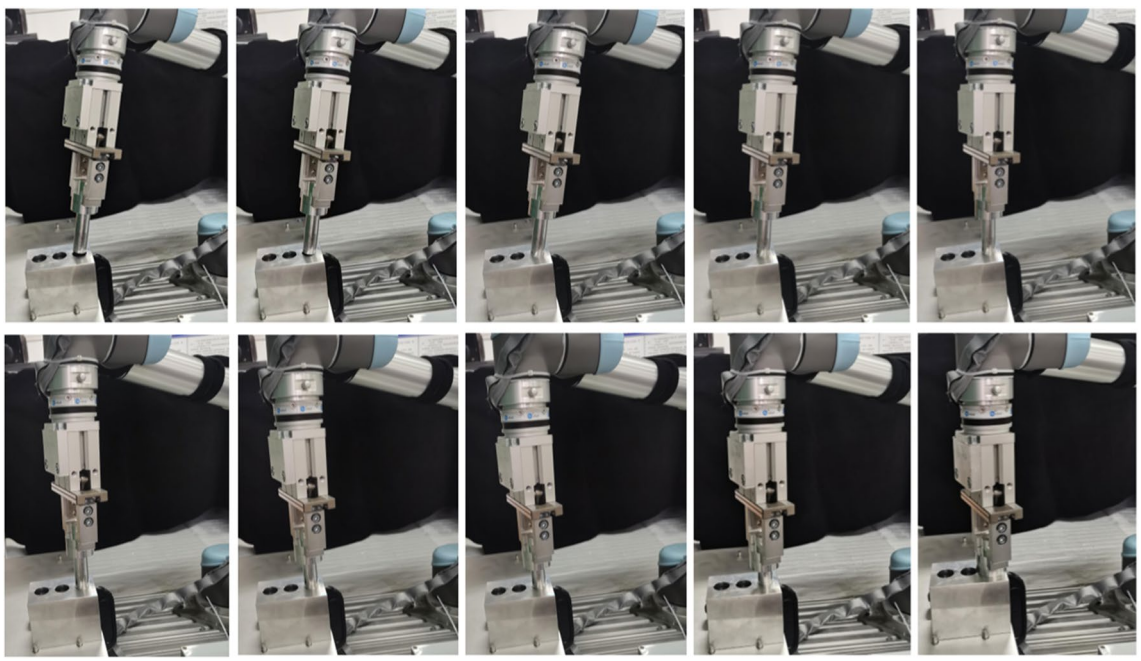


Fig. 7 The peg-in-hole assembly insertion process

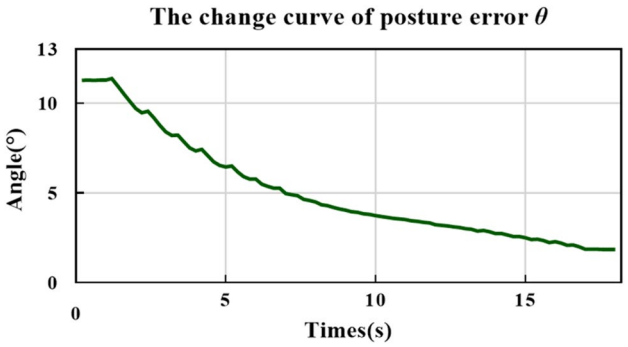


Fig. 8 The change curve of posture error θ in the insertion process

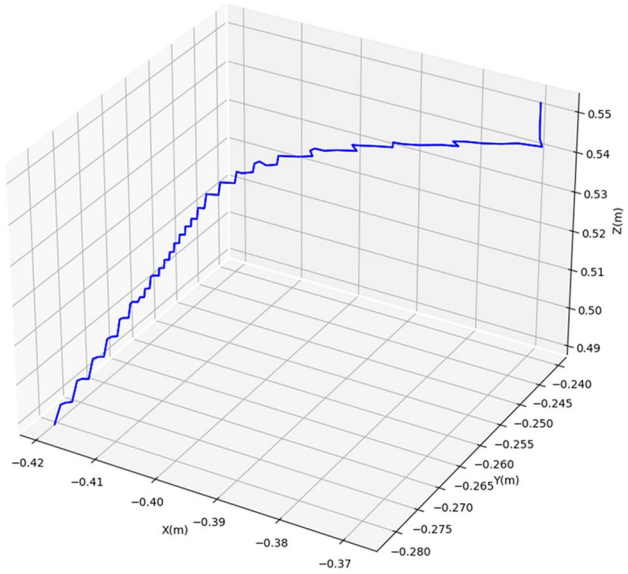


Fig. 9 The assembly trajectory of the robotic arm end

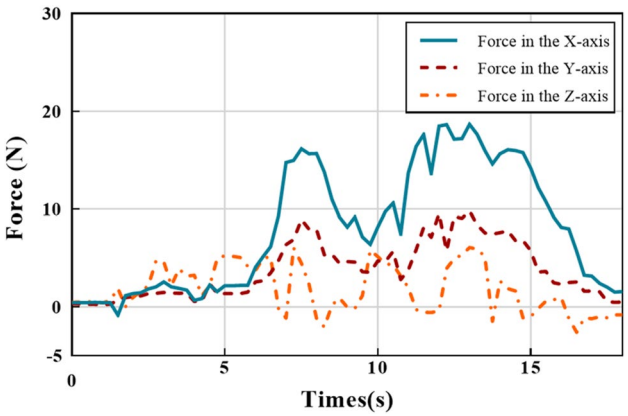


Fig. 10 The change of contact force in the insertion process

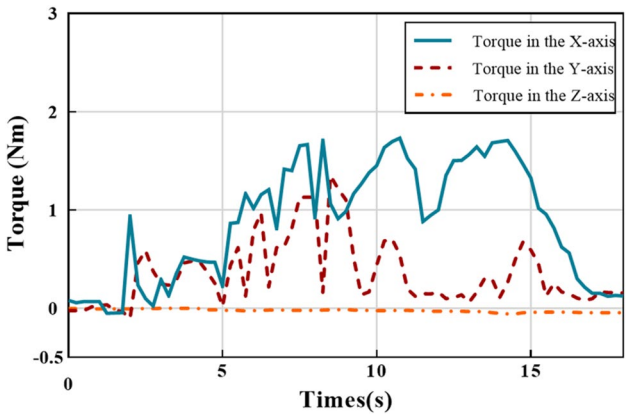


Fig. 11 The change of contact torque in the insertion process

Table 5 The time consumed by the three contact states

Hole diameter(mm)	Clearance(mm)	Number of tests	Average time (s)	Success rate
23.40	0.40 mm	50	18.60	96%
23.80	0.80 mm	50	16.90	100%
24.20	1.20 mm	50	15.40	100%

in the X and Y directions is limited to about 0.5 Nm. The pose of the peg is adjusted according to the force information. At around 4 s, the process transitions to a three-point contact state. This transitional stage has a short duration, and the torque generated in this state is relatively small, requiring a smaller impedance parameter for posture adjustment. Then, at around 4.7 s, the process enters the two-point contact state. In this state, the lateral error is minimal, and the space for adjusting the peg's position becomes limited. Consequently, the force information in this state is relatively complex and requires the most time. Finally, it takes about 14 s to successfully insert the peg into the hole.

Based on the experimental process and method, the performance of the CRL insertion phase is investigated under three different conditions with clearances of 0.40 mm, 0.80 mm, and 1.20 mm. The test results are presented in Table 5. 50 tests are conducted under each clearance condition, with an average completion time of 20 s. When the clearance is 0.40 mm, the success rate reaches 96%. Under clearances of 0.80 and 1.20 mm, the success rate of assembly is 100%. The above experimental results indicate that the CRL method proposed in this study can accomplish the peg-in-hole insertion task successfully under different clearances. It demonstrates a certain level of robustness for different clearances and effectively reduces the assembly force during the insertion process.

In reference [15], a peg-in-hole insertion experiment was conducted using reinforcement learning, and a variable stiffness impedance controller was trained with a peg-hole clearance of 0.4 mm. The entire training process took about 9 h, and the reward curve converged after about 1250 episodes. In the insertion phase experiment, a posture error of 8° was maintained, while the assembly contact force remained within 30 N. In our method, under the same clearance of 0.4 mm, the complete training process takes about 7 h, with the reward curve converging after about 420 episodes. Additionally, the overall assembly contact force is kept within 20 N, despite an initial posture error of about 10°, which is larger than that of the reference [15]. These experiments demonstrate that the proposed CRL method can efficiently generate active compliance control in the insertion phase of the peg-in-hole assembly task.

5 Conclusion

This paper proposes a novel strategy for peg-in-hole assembly, which utilizes two Deep Q-Networks combined with reinforcement learning (CRL) to achieve variable impedance control for high-precision peg-in-hole assembly tasks. The proposed method divides the insertion phase of the peg-in-hole assembly into three contact states. Unlike traditional assembly strategies that require much prior knowledge for establishing contact models, CRL does not rely on such information. The peg-in-hole insertion experiments were conducted under three different clearances, with the task being completed within 20 s. When the peg-in-hole clearance is 0.40 mm, the contact force/torque in each direction is controlled within 20 N and 2 N·m, respectively, resulting in an assembly success rate of 96%. Moreover, the assembly success rate reaches 100% when the peg-in-hole clearance is 0.80 and 1.20 mm. In the future, a more complete parameter optimization strategy will be developed to improve the applicability and training efficiency of the model, further expanding its potential applications.

Author contributions Conception of project, Chengjun Chen; execution, Chengjun Chen and Chenxu Zhang.; manuscript preparation, Chenxu Zhang and Yong Pan.; writing of the first draft, Chengjun Chen and Chenxu Zhang. All authors have read and agreed to the published version of the manuscript. All authors have read and agreed to the published version of the manuscript.

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Data availability The datasets generated or analyzed during this study are available from the corresponding author on reasonable request.

Declarations

Ethical and informed consent for data used All data used in this paper conforms to the Ethical and informed consent specification.

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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